Submission PHYS1-PS1-SUB06

Question PHYS1-PS1-Q01
notation changes halfway:
1. $v \approx v\blacksquare + at \approx 3+2(5) \approx 13$ m/s
2. $\Delta x \approx v\blacksquare t + \frac{1}{2}at^2 \approx 3(5)+\frac{1}{2}(2)(5)^2 \approx 40$ m
marks, subscripts, and units are mixed up
result written as: v=13 m/s; displacement=40 m (please interpret)

Question PHYS1-PS1-Q02
notation changes halfway:
1. $N \approx mg \approx 29.4$ N; friction $\approx \mu N \approx 5.9$ N
2. $\Sigma F \approx 18 - 5.9 \approx 12.1$ N
3. $a \approx \Sigma F/m \approx 4.04$ m/s²
marks, subscripts, and units are mixed up
result written as: a=4.04 m/s²; friction opposes motion (please interpret)

Question PHYS1-PS1-Q03
notation changes halfway:
1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{(2gh)} \approx \sqrt{(2\cdot 9.8\cdot 7)}$
2. $v \approx 11.71$ m/s; mass cancels
marks, subscripts, and units are mixed up
result written as: v=11.71 m/s (please interpret)

Question PHYS1-PS1-Q04
notation changes halfway:
1. p_total,before ≈ p_total,after
2. It applies when external impulse on the system is negligible.
marks, subscripts, and units are mixed up
result written as: total momentum before equals after when external impulse is negligible (please interpret)

Question PHYS1-PS1-Q05
notation changes halfway:
1. Impulse is area under F(t): $J \approx F\Delta t \approx 6(0.5) \approx 3$ N·s.
2. $\Delta p \approx J$, so momentum changes by 3 kg·m/s in the force direction.
marks, subscripts, and units are mixed up
result written as: impulse=3 N·s; momentum changes by 3 kg·m/s (please interpret)

Question PHYS1-PS1-Q06
notation changes halfway:
1. $v \approx v\blacksquare + at \approx 4+4(3) \approx 16$ m/s
2. $\Delta x \approx v\blacksquare t + \frac{1}{2}at^2 \approx 4(3)+\frac{1}{2}(4)(3)^2 \approx 30$ m
marks, subscripts, and units are mixed up
result written as: v=16 m/s; displacement=30 m (please interpret)